Software Test Plan (STP)

Demand Forecasting & Inventory Planning System

Field	Details
Project	Demand Forecasting & Inventory Planning System
Version	1.1
Authors	Team 03 - Rahul Saran Sharma, Vishnu P, Sahana K Golabanvi, K Preksha
Date	12-10-2026
Status	Draft / Planned Test Plan

Revision History

Version	Date	Author	Change Summary
1.0	07-10-2026	Team 03	Initial STP tailored to assigned mini-project.
1.1	12-10-2026	Team 03	Consistency update with SRS/SAD v1.1; added DFIP-F-023 / TC-CFG-01 and finalized forecasting, planning and CSV test details.

1. Introduction

1.1 Purpose

This Software Test Plan defines the testing objectives, scope, strategy, environment, responsibilities, schedule, traceability, metrics, and acceptance criteria for the Demand Forecasting & Inventory Planning System. The plan is derived from the Team 03 SRS and SAD and is intended to verify that the implemented system satisfies its functional, non-functional, and security requirements.

1.2 Scope

Testing covers authentication and role-based access, product and inventory management, historical sales data entry/import, demand forecasting, reorder recommendations, stockout/overstock risk identification, dashboard/alerts, reporting/export, user administration, planning-parameter configuration, audit logging, performance, reliability, usability, maintainability review, and security controls. Direct supplier purchase ordering, payment processing, physical warehouse hardware, and third-party ERP integrations are outside the current mini-project scope.

1.3 References

- Software Requirements Specification (SRS), Demand Forecasting & Inventory Planning System, Version 1.1.
- Software Architecture and Design Specification (SAD), Demand Forecasting & Inventory Planning System, Version 1.1.
- Course-provided Software Test Plan template for Software Engineering.

1.4 Definitions

Term	Meaning
DFIP	Demand Forecasting & Inventory Planning
SKU	Stock Keeping Unit
STP	Software Test Plan
SRS	Software Requirements Specification

Term	Meaning
SAD	Software Architecture and Design Specification
RTM	Requirements Traceability Matrix
UAT	User Acceptance Testing
CSV	Comma-Separated Values

2. Test Items

- Authentication and Authorization Module
- Product Management Module
- Inventory Management Module
- Sales Data and CSV Import Module
- Demand Forecasting Module
- Inventory Planning and Risk Analysis Module
- Dashboard and Alert Module
- Reporting and CSV Export Module
- Administration, Planning Settings and Audit Module
- REST API / Backend service interfaces
- Relational database persistence and recovery behavior

3. Features to be Tested

Requirement IDs	Feature Area	Primary Test References
DFIP-F-001 to F-003	User login, role-aware access, logout and session termination.	TC-AUTH-01 to TC-AUTH-03
DFIP-F-004 to F-006	Product/SKU creation and maintenance; inventory storage and validation.	TC-PROD-01, TC-INV-01 to 02
DFIP-F-007 to F-009	Historical sales entry, CSV import, filtering and viewing.	TC-SALES-01 to 03
DFIP-F-010 to F-012	Forecast generation, horizon selection and forecast result storage.	TC-FCST-01 to 03
DFIP-F-013 to F-016	Reorder quantity, stockout/overstock risk and recommendation input review.	TC-PLAN-01 to 04
DFIP-F-017 to F-020	Dashboard, risk alerts, reports and CSV export.	TC-DASH-01 to 02, TC-RPT-01 to 02
DFIP-F-021 to F-023	User/role administration, planning-parameter configuration and audit logging.	TC-ADM-01, TC-AUD-01, TC-CFG-01
DFIP-NF-001 to NF-005	Performance, reliability, usability and maintainability requirements.	TC-PERF-01 to 02, TC-REL-01, TC-UX-01, TC-MAINT-01
DFIP-SR-001 to SR-005	Password protection, authentication, authorization, input validation and HTTPS/TLS support.	TC-SEC-01 to 05

4. Features Not to be Tested

- Automatic placement of purchase orders with external suppliers, because it is outside the SRS scope.
- Payment gateway or banking transactions.
- Physical warehouse devices such as barcode scanners, RFID readers, or automated storage hardware.
- Internal correctness of third-party libraries/frameworks; only the way the project uses them will be tested.
- Large enterprise-scale load beyond the academic-scale performance targets defined in the SRS.

5. Test Approach / Strategy

5.1 Test Levels

Level	Purpose	Examples
Unit Testing	Verify individual functions/classes in isolation.	Reorder calculation, validation rules, forecast utility functions.
Integration Testing	Verify communication between components.	Web/API to database, backend to forecasting engine, import to storage.
System Testing	Verify complete end-to-end user flows.	Login -> import sales -> forecast -> reorder -> report.
Acceptance Testing (UAT)	Confirm system meets SRS acceptance criteria from user/evaluator perspective.	Inventory planner and administrator scenarios.

5.2 Test Types

- Functional testing: verify each DFIP-F requirement and its acceptance criteria.
- Regression testing: re-run affected tests after fixes or changes.
- Performance testing: validate dashboard/data retrieval and forecast response-time targets.
- Reliability testing: verify committed data survives normal application restart.
- Usability testing: check screen readability, navigation, and validation messages.
- Security testing: validate credential storage, protected resources, server-side authorization, input validation, and HTTPS/TLS configuration.
- Data validation testing: verify malformed CSV rows, negative values, missing fields, invalid dates, unsupported forecast horizons, insufficient history, and inconsistent ranges are safely handled.

5.3 Entry and Exit Criteria

Criteria	Condition
Entry Criteria	Stable build available; database initialized; representative test data prepared; required services running; SRS/SAD baselined for the test cycle.
Exit Criteria	100% of planned high-priority tests executed; no open critical defects; all high-priority SRS acceptance criteria satisfied; failed medium-priority tests documented with agreed disposition; RTM updated.

5.4 Security Validation

- Verify passwords are not stored in plaintext (DFIP-SR-001).
- Verify unauthenticated users cannot access protected pages/APIs (DFIP-SR-002).
- Verify non-admin users cannot bypass server-side authorization through direct requests (DFIP-SR-003).
- Submit malformed input and malicious-looking strings to verify validation/sanitization without data corruption (DFIP-SR-004).
- Verify HTTPS/TLS is enabled in network deployment and credentials are not sent over plaintext HTTP (DFIP-SR-005).

6. Test Environment

Category	Planned Environment
Client Hardware	Standard laptop/desktop with at least 4 GB RAM.
Client Software	Current Chrome or Microsoft Edge browser.
Application Server	Windows or Linux machine running the project backend and forecasting service.
Database	Relational database used by the implementation; automated tests use an isolated temporary database with tables created by fixtures.

Category	Planned Environment
Application Stack	Web frontend + backend REST APIs + Python-based forecasting/analytics module or equivalent implementation.
Tools	pytest and FastAPI TestClient for automated backend tests; Postman/Swagger for API checks; browser developer tools; Git/GitHub; optional JMeter/Locust for performance tests.
Test Data	Synthetic/dummy products, users/roles, inventory levels, and sales history including normal, increasing, seasonal and sparse demand patterns; forecast histories with at least 8 periods; 1-12 week and 1-6 month horizons; CSV files using sku,date,quantity_sold with YYYY-MM-DD dates, including valid and invalid rows.

7. Test Schedule

The following is the planned documentation and initial test-preparation schedule for the current mini-project milestone. Dates may be updated when the implementation build is available.

Activity	Planned Date	Owner
Finalize SRS/SAD/STP and test-case mapping	07-10-2026 to 12-10-2026	Team 03
Prepare test data and test environment	09-10-2026	Team 03
Review test cases and traceability	10-10-2026	Team 03
Documentation milestone / repository check	12-10-2026	Team 03
Functional and integration execution	During implementation test cycle	Assigned testers/developers
Final regression and UAT	Before final mini-project submission	Team 03 / Evaluator

8. Test Deliverables

- Software Test Plan (this document).
- Requirement-to-test traceability matrix.
- Manual and/or automated test cases.
- Test data and CSV samples.
- Automated test scripts where implemented.
- Test execution evidence/logs/screenshots.
- Defect log and defect status.
- Final Test Summary Report.

9. Roles and Responsibilities

Role	Team Member	Responsibility
Team Coordinator / Documentation	Sahana K Golabanvi	Maintain SRS/SAD/STP consistency, test documentation and traceability.
Developer / Unit Testing	Rahul Saran Sharma	Implement assigned modules and execute unit-level checks.
Developer / Integration Testing	Vishnu P	Support API/database integration and integration test execution.
QA / Functional Testing	K Preksha	Execute functional scenarios, record results and defects.
All Team Members	Team 03	Review requirements, reproduce defects, support regression and acceptance testing.

10. Risks and Mitigation

Risk	Impact	Mitigation
Insufficient or unrealistic historical sales data	Forecast tests may not represent real behavior.	Use a documented synthetic dataset with normal, increasing, seasonal and sparse demand patterns.
Forecast algorithm changes during development	Expected results may change.	Version the algorithm/configuration and update test baselines when changes are approved.
Late integration of frontend, backend and database	End-to-end testing may be delayed.	Test APIs/modules independently first; use stubs/mock data until integration is ready.

Risk	Impact	Mitigation
CSV format inconsistencies	Import tests may fail unpredictably.	Use the fixed import schema sku,date,quantity_sold; dates use YYYY-MM-DD; maintain valid/invalid sample files and row-level validation.
Limited time before submission	Regression coverage may be reduced.	Prioritize High requirements, automate repeatable calculations where practical, and maintain the RTM continuously.

11. Assumptions & Dependencies

- The SRS requirement IDs remain stable for Version 1.1 unless a controlled update is made.
- The project implementation provides a seeded or test database and representative historical sales data.
- The forecasting implementation follows the approved rules: at least 8 periods of valid history; horizons of 1-12 weeks or 1-6 months; Holt-Winters when at least two seasonal cycles are available, otherwise simple exponential smoothing; negative predictions are clipped to zero.
- The chosen database and application runtime are available in the team test environment, and automated test fixtures create the required tables on a fresh test run.
- All team members use the shared team repository for source code, documents, test assets, and version control.

12. Suspension & Resumption Criteria

Action	Criteria
Suspend Testing	Application cannot start; database unavailable for an extended period; build blocks more than 30% of planned test cases; critical data-corruption defect is found; required test dataset is missing.
Resume Testing	Blocking issue is corrected; environment and database are stable; smoke test passes; affected data is restored/reset; revised build is available.

13. Detailed Test Cases & Traceability

The following core test cases directly use the SRS test identifiers. Detailed execution evidence can be maintained in the repository as implementation progresses.

Test ID	Requirement	Scenario / Steps	Expected Result	Priority
TC-AUTH-01	DFIP-F-001	Login with valid credentials; repeat with invalid password.	Valid user obtains an authorized session; invalid credentials are rejected.	High
TC-AUTH-02	DFIP-F-002	Login as non-admin and attempt an administrator-only operation through UI/API.	Operation is denied for unauthorized role.	High
TC-INV-02	DFIP-F-006	Update stock with valid quantity, then negative/non-numeric quantity.	Valid value persists; invalid values are rejected.	High
TC-SALES-02	DFIP-F-008	Import a valid CSV using sku,date,quantity_sold (YYYY-MM-DD), then import files with missing/invalid rows.	Valid rows import correctly; invalid rows are reported without corrupting existing data.	High

Test ID	Requirement	Scenario / Steps	Expected Result	Priority
TC-FCST-01	DFIP-F-010	Select a product with at least 8 valid periods of sales history and request a forecast.	Forecast is generated and displayed/stored; fewer than 8 periods is rejected as insufficient history.	High
TC-FCST-02	DFIP-F-011	Generate forecasts for supported horizons: 1-12 weeks and 1-6 months.	Output period count and period spacing match the selected unit and horizon.	High
TC-PLAN-01	DFIP-F-013	Use known forecast demand, safety stock and current stock; calculate reorder quantity.	reorder_qty = max(0, ceil(forecast_demand + safety_stock - current_stock)).	High
TC-PLAN-02	DFIP-F-014	Use current_stock lower than forecast_demand.	Product is flagged as stockout risk.	High
TC-PLAN-03	DFIP-F-015	Use current_stock greater than overstock_factor x forecast_demand.	Product is flagged as overstock risk.	High
TC-DASH-01	DFIP-F-017	Open dashboard after loading known test data.	Dashboard values match latest stored inventory, forecasts and recommendations.	High
TC-RPT-02	DFIP-F-020	Generate filtered report and export CSV.	CSV downloads and matches displayed report records.	Medium
TC-AUD-01	DFIP-F-022	Perform login and data-changing actions; inspect audit records.	Audit entries include actor, action, timestamp and result.	Medium
TC-CFG-01	DFIP-F-023	Login as administrator; update default safety stock, overstock factor and minimum history; read settings again. Repeat as a non-admin.	Valid administrator changes persist and are returned; non-admin update is rejected.	High

Planning and forecasting oracle rules for v1.1: minimum history = 8 periods; supported horizons = 1-12 weeks or 1-6 months; reorder quantity = max(0, ceil(forecast demand + safety stock - current stock)); stockout if current stock < forecast demand; overstock if current stock > overstock factor x forecast demand.

13.1 Non-Functional and Security Test Cases

Test ID	Requirement	Scenario / Steps	Expected Result	Priority
TC-PERF-01	DFIP-NF-001	Execute representative dashboard/data retrieval requests under normal academic-scale load and record response times.	At least 90% complete within 3 seconds.	High
TC-PERF-02	DFIP-NF-002	Generate a single-product forecast using the defined test dataset and measure runtime.	Forecast completes within 10 seconds.	High
TC-REL-01	DFIP-NF-003	Create/modify committed records, restart application normally, then re-query data.	Committed records remain available and consistent.	High

Test ID	Requirement	Scenario / Steps	Expected Result	Priority
TC-UX-01	DFIP-NF-004	Review main screens at common desktop resolutions and trigger validation errors.	UI remains usable and messages are clear.	Medium
TC-MAINT-01	DFIP-NF-005	Review code/module boundaries and documented interfaces.	Forecasting, inventory and reporting concerns are modular and independently maintainable.	Medium
TC-SEC-01	DFIP-SR-001	Inspect stored user credential representation.	No plaintext passwords are stored.	High
TC-SEC-02	DFIP-SR-002	Request protected page/API without authentication.	Request is rejected or redirected to login.	High
TC-SEC-03	DFIP-SR-003	Send direct admin request while authenticated as non-admin.	Backend rejects the operation.	High
TC-SEC-04	DFIP-SR-004	Submit malformed fields, invalid CSV values and script/SQL-like strings.	Input is rejected/safely handled; no unintended execution/data corruption.	High
TC-SEC-05	DFIP-SR-005	Inspect network deployment endpoint/protocol.	HTTPS/TLS is available and credentials are not transmitted via plaintext HTTP.	Medium

13.2 Requirement-to-Test Traceability Summary

Requirement ID	Mapped Test Case
DFIP-F-001	TC-AUTH-01
DFIP-F-002	TC-AUTH-02
DFIP-F-003	TC-AUTH-03
DFIP-F-004	TC-PROD-01
DFIP-F-005	TC-INV-01
DFIP-F-006	TC-INV-02
DFIP-F-007	TC-SALES-01
DFIP-F-008	TC-SALES-02
DFIP-F-009	TC-SALES-03
DFIP-F-010	TC-FCST-01
DFIP-F-011	TC-FCST-02
DFIP-F-012	TC-FCST-03
DFIP-F-013	TC-PLAN-01
DFIP-F-014	TC-PLAN-02
DFIP-F-015	TC-PLAN-03
DFIP-F-016	TC-PLAN-04
DFIP-F-017	TC-DASH-01
DFIP-F-018	TC-DASH-02
DFIP-F-019	TC-RPT-01
DFIP-F-020	TC-RPT-02
DFIP-F-023	TC-CFG-01
DFIP-F-021	TC-ADM-01
DFIP-F-022	TC-AUD-01
DFIP-NF-001	TC-PERF-01
DFIP-NF-002	TC-PERF-02
DFIP-NF-003	TC-REL-01
DFIP-NF-004	TC-UX-01
DFIP-NF-005	TC-MAINT-01

Requirement ID	Mapped Test Case
DFIP-SR-001	TC-SEC-01
DFIP-SR-002	TC-SEC-02
DFIP-SR-003	TC-SEC-03
DFIP-SR-004	TC-SEC-04
DFIP-SR-005	TC-SEC-05

14. Test Metrics & Reporting

- Test execution percentage = executed test cases / planned test cases x 100.
- Pass percentage = passed test cases / executed test cases x 100.
- Requirement coverage = requirements with at least one executed test / total testable requirements x 100.
- Defect count by severity (Critical, High, Medium, Low).
- Defect closure/retest status.
- Performance results for dashboard/data retrieval and forecast generation.
- Daily/iteration-level execution status during active testing and a final Test Summary Report before submission.

15. Approvals

Role	Name	Signature / Date
QA / Test Lead		
Development Lead		
Team Coordinator		
Course Instructor / Evaluator		